WORKED SOLUTION

STEP I 2002, Question 3

Pure • Inequalities

> **STEP I 2002, Question 3**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/02-s1-q3

The key idea

Call the two square roots $\sqrt{P}$ and $\sqrt{Q}$. The quantities under them are nicely linked: $P + Q = a^2 + b^2$ is constant, and their derivatives are equal and opposite. That makes $\mathrm{d}y/\mathrm{d}\theta$ factorise cleanly, and its sign is controlled by a single trigonometric factor. Once we know the largest and smallest values of y, the double inequality is just "min $\leq y \leq$ max", and the first part tells us the two ends are in the right order.

Solution

The first inequality

Expanding,

$$2a^2 + 2b^2 - (a+b)^2 = a^2 - 2ab + b^2 = (a-b)^2 \geq 0,$$

so $(a+b)^2 \leq 2a^2 + 2b^2$, with equality exactly when $a = b$.

The stationary points

Write $c = \cos\theta$, $s = \sin\theta$ and

$$P = a^2c^2 + b^2s^2, \qquad Q = a^2s^2 + b^2c^2, \qquad y = \sqrt{P} + \sqrt{Q}.$$

Two special cases first. If $a^2 = b^2$, then $P = Q = a^2$ for every θ, so $y = 2|a|$ is constant and every point is (trivially) stationary; we set this aside. If $a = 0$ or $b = 0$ (but not both), P or Q vanishes at some θ and y is not differentiable there; we deal with this at the end. So for now assume $ab \neq 0$ and $a^2 \neq b^2$. Then $P \geq \min(a^2, b^2)(c^2 + s^2) > 0$, and likewise $Q > 0$, so y is differentiable everywhere.

Differentiating,

$$\frac{\mathrm{d}P}{\mathrm{d}\theta} = -2a^2cs + 2b^2sc = (b^2 - a^2)\sin 2\theta, \qquad \frac{\mathrm{d}Q}{\mathrm{d}\theta} = (a^2 - b^2)\sin 2\theta.$$

Hence

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} = \frac{(b^2-a^2)\sin 2\theta}{2\sqrt{P}} - \frac{(b^2-a^2)\sin 2\theta}{2\sqrt{Q}} = \frac{(b^2-a^2)\sin 2\theta\,(\sqrt{Q}-\sqrt{P})}{2\sqrt{PQ}}.$$

To see the sign of $\sqrt{Q}-\sqrt{P}$, rationalise:

$$\sqrt{Q}-\sqrt{P} = \frac{Q-P}{\sqrt{Q}+\sqrt{P}}, \qquad Q-P = (a^2-b^2)(s^2-c^2) = (b^2-a^2)\cos 2\theta.$$

Putting this in, and using $2\sin 2\theta\cos 2\theta = \sin 4\theta$,

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} = \frac{(a^2-b^2)^2\sin 4\theta}{4\sqrt{PQ}\,(\sqrt{P}+\sqrt{Q})}.$$

? **Ask yourself** Why is it worth the trouble of rationalising $\sqrt{Q}-\sqrt{P}$?

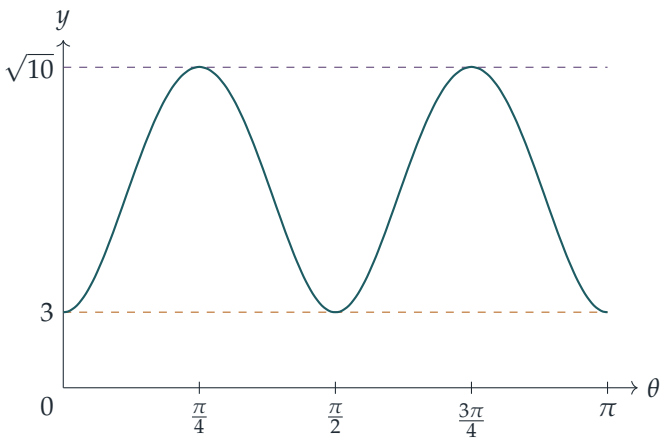
Because now everything except $\sin 4\theta$ is visibly positive: $(a^2-b^2)^2 > 0$, and P, Q are positive. So $\mathrm{d}y/\mathrm{d}\theta$ has exactly the sign of $\sin 4\theta$, which tells us both where the stationary points are and what kind they are.

The stationary points are where $\sin 4\theta = 0$, that is $\theta = \frac{1}{4}n\pi$ for integer n.

- When n is even, $\theta = \frac{1}{2}k\pi$, so $\{c^2, s^2\} = \{1, 0\}$ and $\{P, Q\} = \{a^2, b^2\}$. Here $y = |a| + |b|$ (remember $\sqrt{a^2} = |a|$).
- When n is odd, $\theta = \frac{1}{4}\pi + \frac{1}{2}k\pi$, so $c^2 = s^2 = \frac{1}{2}$ and $P = Q = \frac{1}{2}(a^2+b^2)$. Here $y = 2\sqrt{\frac{1}{2}(a^2+b^2)} = \sqrt{2a^2+2b^2}$.

Nature. As θ increases through $\frac{1}{2}k\pi$, $\sin 4\theta$ changes from negative to positive, so y goes from decreasing to increasing: these points are **minima**. As θ increases through $\frac{1}{4}\pi + \frac{1}{2}k\pi$, $\sin 4\theta$ changes from positive to negative, so these points are **maxima**.

The picture below shows $a = 2$, $b = 1$, with the vertical scale stretched: y oscillates between 3 and $\sqrt{10} \approx 3.162$, with period $\frac{1}{2}\pi$.



The final inequality

Replacing θ by $\theta + \frac{1}{2}\pi$ swaps c^2 and s^2, hence swaps P and Q, so y has period $\frac{1}{2}\pi$. On one period $[0, \frac{1}{2}\pi]$, y increases on $[0, \frac{1}{4}\pi]$ and decreases on $[\frac{1}{4}\pi, \frac{1}{2}\pi]$ (sign of $\sin 4\theta$), so its least value is $|a| + |b|$, at the ends, and its greatest is $\sqrt{2a^2 + 2b^2}$, at $\frac{1}{4}\pi$. By periodicity these are the least and greatest values for all θ, so

$$|a| + |b| \;\le\; y \;\le\; \sqrt{2a^2 + 2b^2}.$$

These two bounds are consistent: applying the first part to $|a|$ and $|b|$ gives $(|a| + |b|)^2 \le 2a^2 + 2b^2$, so the lower bound really does not exceed the upper one.

The cases set aside. If $a^2 = b^2$, then $y = 2|a|$, while $|a| + |b| = 2|a|$ and $\sqrt{2a^2 + 2b^2} = \sqrt{4a^2} = 2|a|$, so the inequality holds with equality throughout. If $b = 0$, then $y = |a|(|\cos\theta| + |\sin\theta|)$. Now $|\cos\theta| + |\sin\theta| \ge \cos^2\theta + \sin^2\theta = 1$ since each term is at most 1, and by the first part $(|\cos\theta| + |\sin\theta|)^2 \le 2$; so $|a| \le y \le \sqrt{2}\,|a|$, which is the required inequality. The case $a = 0$ is the same with a and b swapped, and $a = b = 0$ is trivial. This proves the result for all a, b and θ. ■

Checks

- With $a = 2$, $b = 1$: $y(0) = 2 + 1 = 3$ and $y(\frac{1}{4}\pi) = 2\sqrt{5/2} = \sqrt{10} \approx 3.1623$; a numerical scan of y over many θ stays between these.
- The derivative formula agrees with a finite-difference estimate of $\mathrm{d}y/\mathrm{d}\theta$ to about 10^{-9} at random a, b, θ.
- With $a = b$ both bounds collapse to $2|a|$, matching the constant y.

Common slips

- Writing $\sqrt{a^2} = a$: the minimum value is $|a| + |b|$, which is why moduli appear in the answer.
- Stopping at $\sin 2\theta = 0$ and missing the stationary points at odd multiples of $\frac{1}{4}\pi$, where $\sqrt{P} = \sqrt{Q}$.
- Calling points maxima or minima without a reason; the sign of $\sin 4\theta$ gives one in a line.

Beyond STEP

Geometrically, $\sqrt{P}$ is the distance from the centre of the ellipse $x = a\cos\phi$, $y = b\sin\phi$ to a point on it, and $\sqrt{Q}$ is the distance to the point a quarter-turn further round in parameter, so the question is about how these two "conjugate" radii share out a fixed total $a^2 + b^2$. The two bounds are also the ℓ^1 and (scaled) ℓ^2 norms of the vector (a, b),

and comparing norms like this is a routine tool in first-year analysis and linear algebra. For more on using derivatives to prove inequalities, see Abbott, *Understanding Analysis*.

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